

PROJECT 31A — DATABASE SCHEMA BIBLE (DOCUMENT 3)

Version 1.0.0

**Classification: CONFIDENTIAL — FOR INTERNAL
DEVELOPMENT AND DUE DILIGENCE ONLY**

1. DATABASE ENGINE & CONNECTION CONFIGURATION

Project 31A utilizes **PostgreSQL 16** as its core relational database engine. All database interactions from the FastAPI backend are asynchronous, utilizing the **asyncpg** driver combined with **SQLAlchemy 2.x async ORM**.

1.1 Connection Pooling and Lifetime Config

Database connections are managed via SQLAlchemy's `async_sessionmaker` and `create_async_engine`. The configuration is defined in `backend/src/seo_platform/config.py` and instantiated in `backend/src/seo_platform/core/database.py`.

- **Pool Size** (`settings.database.pool_size`): Default is set to `20` concurrent connections per worker process.
- **Max Overflow** (`settings.database.max_overflow`): Allows up to `10` additional connections beyond pool size under peak load.
- **Pool Recycle** (`settings.database.pool_recycle`): Connections are recycled every `1800` seconds (30 minutes) to prevent stale connections and leak accumulation.
- **Pool Pre-Ping**: Enforced to check connection viability (`SELECT 1`) before checking out from the pool.

1.2 Multi-Tenant Session Scoping

The platform enforces absolute isolation of tenant data at the query level. Sessions are initialized via `get_tenant_session(tenant_id: UUID)` which returns an async session configured with:

1. An automatic query interceptor that appends `WHERE tenant_id = :tenant_id` to all SELECT, UPDATE, and DELETE queries targeting tenant-specific models.
 2. Safe execution parameters ensuring no cross-tenant leakage is mathematically possible.
-

2. BASE DECLARATIVE MIXINS

To ensure model schema uniformity, all models inherit from `Base` and utilize a set of reusable SQLAlchemy mixins located in `backend/src/seo_platform/models/base.py`.

2.1 UUIDPrimaryKeyMixin

Provides a standardized UUID primary key.

```
class UUIDPrimaryKeyMixin:
    id = Column(
        UUID(as_uuid=True),
        primary_key=True,
        default=uuid.uuid4,
        server_default=text("gen_random_uuid()")
    )
```

- **Engine-level default:** `gen_random_uuid()` from PostgreSQL `pgcrypto` extension.
- **App-level default:** `uuid.uuid4` if no primary key is explicitly provided during instantiation.

2.2 TenantMixin

Applies multi-tenancy requirements to models.

```
class TenantMixin:
    tenant_id = Column(
        UUID(as_uuid=True),
        ForeignKey("tenants.id", ondelete="CASCADE"),
        nullable=False,
        index=True
    )
```

- **Foreign Key constraint:** Cascade delete is enforced. If a Tenant is deleted, all child records across all tables are purged automatically.
- **Indexing:** The `tenant_id` column is indexed by default to optimize tenant-scoped queries.

2.3 TimestampMixin

Adds creation and modification tracking.

```
class TimestampMixin:
    created_at = Column(
        DateTime(timezone=True),
        nullable=False,
        server_default=func.now()
    )
    updated_at = Column(
        DateTime(timezone=True),
        nullable=False,
        server_default=func.now(),
        onupdate=func.now()
    )
```

- Timestamps are timezone-aware, standardizing on UTC inside the database.

3. MULTI-TENANT ISOLATION STRATEGY & RLS

Data isolation is guaranteed through a defense-in-depth approach combining application-level scoping and database-level security.

3.1 Session-Scoping and *gettenant*session

All API route handlers must access the database via the `get_tenant_session(tenant_id)` context manager. This manager validates the request's extracted `tenant_id` against the JWT claims before initiating database operations.

3.2 Row-Level Security (RLS)

PostgreSQL Row-Level Security is enabled on all tables containing a `tenant_id` column. For each table, the following policy is applied during migrations:

```
ALTER TABLE table_name ENABLE ROW LEVEL SECURITY;
CREATE POLICY tenant_isolation_policy ON table_name
    USING (tenant_id = current_setting('app.current_tenant_id')::uuid);
```

During session initialization, the async connection executes: `SET LOCAL app.current_tenant_id = 'uuid';` This guarantees that even if a developer forgets to apply a tenant filter in SQLAlchemy, the PostgreSQL engine will refuse to return or modify any rows belonging to other tenants.

4. COMPLETE TABLE CATALOGUE

4.1 Table: `tenants`

Stores details of the tenant organizations subscribing to the platform.

Column Name	Data Type	Constraints / Attributes	Purpose
<code>id</code>	<code>UUID</code>	Primary Key, <code>server_default=gen_random_uuid()</code>	Unique identifier for the tenant.
<code>slug</code>	<code>VARCHAR(100)</code>	Unique, Indexed, Not Null	URL slug identifier (e.g., <code>acme-corp</code>).
<code>name</code>	<code>VARCHAR(255)</code>	Not Null	Display name of the tenant.
<code>plan</code>	<code>tenant_plan</code>	Enum (starter, growth, enterprise), Not Null	Subscription tier governing limits.
<code>settings</code>	<code>JSONB</code>	Default: <code>{}</code>	Custom tenant configurations.
<code>suspended_at</code>	<code>TIMESTAMP WITH TIME ZONE</code>	Nullable	Timestamp when tenant access was revoked.
<code>created_at</code>	<code>TIMESTAMP WITH TIME ZONE</code>	Not Null, Default: <code>NOW()</code>	Audit timestamp.
<code>updated_at</code>	<code>TIMESTAMP WITH TIME ZONE</code>	Not Null, Default: <code>NOW()</code>	Audit timestamp.

4.2 Table: `users`

Contains user identity records associated with a tenant.

Column Name	Data Type	Constraints / Attributes	Purpose
<code>id</code>	<code>UUID</code>	Primary Key	Unique user identifier.
<code>tenant_id</code>	<code>UUID</code>	Foreign Key (tenants.id) ON DELETE CASCADE	Associated tenant.
<code>external_id</code>	<code>VARCHAR(255)</code>	Unique, Not Null	External identity provider ID (Clerk/Auth0).
<code>email</code>	<code>VARCHAR(255)</code>	Not Null	User email address.
<code>name</code>	<code>VARCHAR(255)</code>	Not Null	User full name.
<code>role</code>	<code>user_role</code>	Enum, Not Null	Role (<i>superadmin</i> , <i>tenantadmin</i> , <i>manager</i> , etc.).
<code>is_active</code>	<code>BOOLEAN</code>	Default: <code>True</code>	User activation status.
<code>permissions</code>	<code>JSONB[]</code>	Default: <code>[]</code>	Explicit array of granted action overrides.
<code>last_login_at</code>	<code>TIMESTAMP WITH TIME ZONE</code>	Nullable	Tracker for activity.
<code>created_at</code>	<code>TIMESTAMP WITH TIME ZONE</code>	Not Null	Audit timestamp.
<code>updated_at</code>	<code>TIMESTAMP WITH TIME ZONE</code>	Not Null	Audit timestamp.

4.3 Table: `clients`

Represents the business entities (e.g., websites or projects) managed by a tenant.

Column Name	Data Type	Constraints / Attributes	Purpose
id	UUID	Primary Key	Unique identifier.
tenant_id	UUID	Foreign Key (tenants.id) ON DELETE CASCADE	Owner tenant.
name	VARCHAR(255)	Not Null	Display name of the client.
domain	VARCHAR(255)	Not Null	Canonical domain (e.g. clientwebsite.com).
niche	VARCHAR(100)	Nullable	Market industry category.
geo_focus	JSONB	Default: []	Target regions/countries.
business_type	business_type	Enum, Nullable	B2B, B2C, eCommerce, SaaS, etc.
onboarding_status	onboarding_status	Enum, Default: pending	Onboarding progress.
status	client_status	Enum, Default: active	Active state of the client.
archived_at	TIMESTAMP WITH TIME ZONE	Nullable	When the client was archived.
profile_data	JSONB	Default: {}	Custom SEO profile and target description.
competitors	JSONB	Default: []	List of target competitor domains.
created_at	TIMESTAMP WITH TIME ZONE	Not Null	Audit timestamp.
updated_at	TIMESTAMP WITH TIME ZONE	Not Null	Audit timestamp.

Constraints: Unique constraint on (tenant_id, domain) .

4.4 Table: backlink_campaigns

Stores the parameters and operational progress of backlink acquisition campaigns.

Column Name	Data Type	Constraints / Attributes	Purpose
<code>id</code>	<code>UUID</code>	Primary Key	Unique campaign ID.
<code>tenant_id</code>	<code>UUID</code>	Foreign Key (tenants.id)	Tenant owner.
<code>client_id</code>	<code>UUID</code>	Foreign Key (clients.id) ON DELETE CASCADE	Associated client.
<code>name</code>	<code>VARCHAR(255)</code>	Not Null	Campaign display name.
<code>campaign_type</code>	<code>campaign_type</code>	Enum, Not Null	<i>guestpost</i> , <i>nicheedit</i> , <i>skyscraper</i> , etc.
<code>status</code>	<code>campaign_status</code>	Enum, Default: <code>draft</code>	Operational state of campaign.
<code>target_link_count</code>	<code>INTEGER</code>	Default: <code>10</code> , Not Null	Number of target links to acquire.
<code>acquired_link_count</code>	<code>INTEGER</code>	Default: <code>0</code> , Not Null	Verified live links acquired.
<code>total_prospects</code>	<code>INTEGER</code>	Default: <code>0</code> , Not Null	Number of prospects discovered.
<code>total_emails_sent</code>	<code>INTEGER</code>	Default: <code>0</code> , Not Null	Total emails sent to targets.
<code>reply_rate</code>	<code>DOUBLE PRECISION</code>	Default: <code>0.0</code> , Not Null	Percentage of prospects that replied.
<code>acquisition_rate</code>	<code>DOUBLE PRECISION</code>	Default: <code>0.0</code> , Not Null	Percentage of prospects resulting in live links.
<code>health_score</code>	<code>DOUBLE PRECISION</code>	Default: <code>0.0</code> , Not Null	Calculated SRE/Business health score.
<code>config</code>	<code>JSONB</code>	Default: <code>{}</code>	Target criteria, parameters, exclusion list.
<code>workflow_run_id</code>	<code>VARCHAR(255)</code>	Nullable	Running Temporal Workflow Run ID.
<code>started_at</code>	<code>TIMESTAMP WITH TIME ZONE</code>	Nullable	Campaign execution start timestamp.
<code>completed_at</code>	<code>TIMESTAMP WITH TIME ZONE</code>	Nullable	Campaign closure timestamp.
<code>created_at</code>	<code>TIMESTAMP WITH TIME ZONE</code>	Not Null	Audit timestamp.
<code>updated_at</code>	<code>TIMESTAMP WITH TIME ZONE</code>	Not Null	Audit timestamp.

4.5 Table: `backlink_prospects`

Discovered target URLs under evaluation for outreach.

Column Name	Data Type	Constraints / Attributes	Purpose
id	UUID	Primary Key	Unique prospect ID.
tenant_id	UUID	Foreign Key (tenants.id)	Tenant owner.
campaign_id	UUID	Foreign Key (backlink_campaigns.id) ON DELETE CASCADE	Campaign container.
domain	VARCHAR(255)	Indexed, Not Null	Root domain of target.
url	VARCHAR(2048)	Not Null	Target page URL containing the link opportunity.
status	prospect_status	Enum, Default: new	State in the pipeline (scored, approved, etc.).
domain_authority	DOUBLE PRECISION	Not Null, Default: 0.0	Target Domain Rating / Authority.
relevance_score	DOUBLE PRECISION	Not Null, Default: 0.0	Semantic relevance to client.
spam_score	DOUBLE PRECISION	Not Null, Default: 0.0	Algorithmic spam risk.
traffic_score	DOUBLE PRECISION	Not Null, Default: 0.0	Estimated monthly search traffic.
composite_score	DOUBLE PRECISION	Not Null, Default: 0.0	Combined AI prioritization score.
confidence	DOUBLE PRECISION	Not Null, Default: 0.0	Confidence in scoring criteria.
contact_name	VARCHAR(255)	Nullable	Scraped contact person name.
contact_email	VARCHAR(255)	Nullable	Target email address.
contact_source	VARCHAR(255)	Nullable	Source from which contact was scraped (e.g. Hunter).
email_verification_status	VARCHAR(50)	Default: 'unverified'	deliverable, undeliverable, risky.
email_verification_result	JSONB	Default: {}	Raw Hunter verification payload.
scoring_rationale	JSONB	Default: {}	Explanatory metrics breakdown.
page_data	JSONB	Default: {}	Raw page scrapings, meta tags, and text density.

Column Name	Data Type	Constraints / Attributes	Purpose
<code>created_at</code>	<code>TIMESTAMP WITH TIME ZONE</code>	Not Null	Audit timestamp.
<code>updated_at</code>	<code>TIMESTAMP WITH TIME ZONE</code>	Not Null	Audit timestamp.

Constraints: Unique constraint on `(tenant_id, campaign_id, domain)`.

4.6 Table: `outreach_threads`

Outbox and reply tracking thread for email campaigns.

Column Name	Data Type	Constraints / Attributes	Purpose
id	UUID	Primary Key	Unique thread ID.
tenant_id	UUID	Foreign Key (tenants.id)	Tenant owner.
campaign_id	UUID	Foreign Key (backlink_campaigns.id)	Associated campaign.
prospect_id	UUID	Foreign Key (backlink_prospects.id) ON DELETE CASCADE	Associated target prospect.
status	thread_status	Enum, Default: <code>draft</code>	Thread state (sent, replied, unsubscribed).
from_email	VARCHAR(255)	Not Null	Sender inbox address.
to_email	VARCHAR(255)	Not Null	Destination address.
subject	VARCHAR(500)	Not Null	Email subject line.
body_html	TEXT	Not Null	Rendered HTML email body.
follow_up_count	INTEGER	Default: <code>0</code> , Not Null	Number of follow-ups sent.
max_follow_ups	INTEGER	Default: <code>3</code> , Not Null	Maximum retry follow-ups.
provider	VARCHAR(50)	Default: <code>'sendgrid'</code>	Platform (sendgrid, resend, mailgun).
provider_message_id	VARCHAR(255)	Nullable	Remote API returned identifier.
sent_at	TIMESTAMP WITH TIME ZONE	Nullable	Initial dispatch timestamp.
opened_at	TIMESTAMP WITH TIME ZONE	Nullable	Tracking pixel trigger timestamp.
replied_at	TIMESTAMP WITH TIME ZONE	Nullable	Inbound webhook trigger timestamp.
confidence_score	DOUBLE PRECISION	Nullable	AI confidence in recipient suitability.
ai_personalization	JSONB	Default: <code>{}</code>	Extracted variables used to construct text.
created_at	TIMESTAMP WITH TIME ZONE	Not Null	Audit timestamp.
updated_at	TIMESTAMP WITH TIME ZONE	Not Null	Audit timestamp.

4.7 Table: `acquired_links`

Tracks successfully completed backlinks and ongoing verification checks.

Column Name	Data Type	Constraints / Attributes	Purpose
id	UUID	Primary Key	Link ID.
tenant_id	UUID	Foreign Key (tenants.id)	Tenant owner.
campaign_id	UUID	Foreign Key (backlink_campaigns.id)	Associated campaign.
prospect_id	UUID	Foreign Key (backlink_prospects.id) ON DELETE SET NULL	Reference back to prospect.
source_url	VARCHAR(2048)	Not Null	Page containing link (prospect website).
target_url	VARCHAR(2048)	Not Null	Target destination page (client website).
anchor_text	VARCHAR(500)	Not Null	HTML anchor text matching link.
link_type	VARCHAR(50)	Default: 'dofollow', Not Null	dofollow, nofollow, sponsored, ugc.
status	link_status	Enum, Default: pending_verification	State of backlink (verified_live, removed).
domain_authority_at_acquisition	DOUBLE PRECISION	Not Null	Domain authority value when link went live.
first_verified_at	TIMESTAMP WITH TIME ZONE	Nullable	Date first found by scraper.
last_checked_at	TIMESTAMP WITH TIME ZONE	Nullable	Date of last verification cron job execution.
check_count	INTEGER	Default: 0, Not Null	Lifetime cron check iterations.
verification_history	JSONB	Default: []	Array of verification check results.
last_http_status	INTEGER	Nullable	HTTP status code returned by target server.
last_response_time_ms	DOUBLE PRECISION	Nullable	Speed of target page response.
last_checked_redirect_chain	JSONB	Nullable	Redirect paths resolved.
last_match_anchor	VARCHAR(500)	Nullable	Actually parsed anchor text.
last_match_rel	VARCHAR(255)	Nullable	Actually parsed rel attributes.
last_match_position	INTEGER	Nullable	Index position in HTML DOM tree.

Column Name	Data Type	Constraints / Attributes	Purpose
last_error	TEXT	Nullable	Scraper extraction error logs.
created_at	TIMESTAMP WITH TIME ZONE	Not Null	Audit timestamp.
updated_at	TIMESTAMP WITH TIME ZONE	Not Null	Audit timestamp.

4.8 Table: audit_log

Append-only history ledger tracking user, system, and agent activity.

Column Name	Data Type	Constraints / Attributes	Purpose
id	BIGSERIAL	Primary Key	Auto-incrementing identifier.
tenant_id	UUID	Foreign Key (tenants.id)	Associated tenant context.
event_type	VARCHAR(100)	Indexed, Not Null	e.g. campaign_launched, outreach_approved.
entity_type	VARCHAR(100)	Not Null	e.g. backlink_campaigns, outreach_threads.
entity_id	UUID	Not Null	Target record UUID.
actor_type	VARCHAR(50)	Not Null	actor: user, system, or agent.
actor_id	VARCHAR(255)	Not Null	Identity of actor (Clerk ID, workflow name).
before_state	JSONB	Nullable	JSON representation of row prior to mutation.
after_state	JSONB	Nullable	JSON representation of row after mutation.
metadata	JSONB	Default: {}	Contextual info (IP, request path, timing).
created_at	TIMESTAMP WITH TIME ZONE	Not Null, Default: NOW()	Event timestamp.

Critical Architecture: This table is structurally **APPEND-ONLY**. Any attempt to modify or delete rows from audit_log triggers an immediate alert and is prevented by a custom PostgreSQL trigger. It is configured for monthly partitioning by created_at.

4.9 Table: `workflow_events`

Ledger supporting event-sourcing architecture.

Column Name	Data Type	Constraints / Attributes	Purpose
<code>id</code>	<code>BIGSERIAL</code>	Primary Key	Sequential ID.
<code>stream_id</code>	<code>UUID</code>	Indexed, Not Null	UUID of aggregate root being tracked.
<code>stream_type</code>	<code>VARCHAR(100)</code>	Not Null	Aggregate namespace (e.g. <code>Campaign</code>).
<code>tenant_id</code>	<code>UUID</code>	Indexed, Not Null	Associated tenant.
<code>sequence_number</code>	<code>INTEGER</code>	Not Null	Sequential index for the specific stream ID.
<code>event_type</code>	<code>VARCHAR(100)</code>	Not Null	Event class identifier.
<code>event_data</code>	<code>JSONB</code>	Not Null	Complete domain event payload.
<code>metadata</code>	<code>JSONB</code>	Default: <code>{}</code>	Actor context, event schema version.
<code>created_at</code>	<code>TIMESTAMP WITH TIME ZONE</code>	Not Null, Default: <code>NOW()</code>	Event ingestion timestamp.

Constraints: Unique constraint on `(stream_id, sequence_number)`. This prevents concurrency issues and race conditions.

4.10 Table: `keywords`

Discovered and monitored keyword opportunities.

Column Name	Data Type	Constraints / Attributes	Purpose
id	UUID	Primary Key	Unique keyword ID.
tenant_id	UUID	Foreign Key (tenants.id)	Tenant owner.
client_id	UUID	Foreign Key (clients.id) ON DELETE CASCADE	Associated client website.
cluster_id	UUID	Foreign Key (keyword_clusters.id) ON DELETE SET NULL	Grouping cluster categorization.
keyword	VARCHAR(500)	Not Null	String phrase.
search_volume	INTEGER	Default: 0, Not Null	Estimated monthly searches.
difficulty	DOUBLE PRECISION	Default: 0.0, Not Null	Difficulty to rank (0-100 scale).
cpc	DOUBLE PRECISION	Default: 0.0, Not Null	Cost-Per-Click in USD.
competition	DOUBLE PRECISION	Default: 0.0, Not Null	Ad competition density.
intent	search_intent	Enum, Nullable	commercial, transactional, informational.
serp_features	JSONB	Default: []	Search page elements present.
enrichment_data	JSONB	Default: {}	Competitor ranks, ranking URLs.
is_seed	BOOLEAN	Default: False	True if keyword was manually input.
embedding_vector_id	VARCHAR(255)	Nullable	ID of associated vector record in Qdrant.
created_at	TIMESTAMP WITH TIME ZONE	Not Null	Audit timestamp.
updated_at	TIMESTAMP WITH TIME ZONE	Not Null	Audit timestamp.

Constraints: Unique constraint on (tenant_id, client_id, keyword) .

5. ENUM REGISTRY

All enums are created as native PostgreSQL enum types. Database-side values must match python enum classes exactly.

5.1 `tenant_plan`

- Values: `starter`, `growth`, `enterprise`
- Governing settings: limits on campaign execution speeds, storage, and API quotas.

5.2 `user_role`

- Values: `super_admin`, `tenant_admin`, `manager`, `seo_analyst`, `outreach_specialist`, `report_analyst`, `client`

5.3 `client_status`

- Values: `active`, `archived`

5.4 `onboarding_status`

- Values: `pending`, `collecting`, `validating`, `ai_enrichment`, `awaiting_approval`, `complete`, `failed_validation`, `ai_failed`, `rejected`

5.5 `business_type`

- Values: `B2B`, `B2C`, `B2B+B2C`, `Marketplace`, `Agency`, `local`, `national`, `ecommerce`, `saas`, `publisher`

5.6 `campaign_type`

- Values: `guest_post`, `resource_page`, `niche_edit`, `broken_link`, `skyscraper`, `haro`

5.7 `campaign_status`

- Values: `draft`, `prospecting`, `scoring`, `outreach_prep`, `awaiting_approval`, `active`, `paused`, `monitoring`, `complete`, `cancelled`, `archived`, `failed_no_prospects`, `failed_no_emails_sent`

5.8 `prospect_status`

- Values: `new`, `scored`, `approved`, `rejected`, `outreach_queued`, `contacted`, `replied`, `link_acquired`, `link_lost`, `unresponsive`

5.9 `thread_status`

- Values: `draft`, `queued`, `sent`, `delivered`, `opened`, `replied`, `bounced`, `spam_reported`, `unsubscribed`, `link_acquired`

5.10 `link_status`

- Values: `pending_verification`, `verified_live`, `verified_nofollow`, `removed`, `broken`

5.11 `search_intent`

- Values: `informational`, `navigational`, `commercial`, `transactional`

5.12 `cluster_status`

- Values: `draft`, `pending_approval`, `approved`, `rejected`, `archived`
-

6. ENTITY RELATIONSHIP DIAGRAM

erDiagram

```
tenants ||--o{ users : "has"
tenants ||--o{ clients : "owns"
clients ||--o{ backlink_campaigns : "contains"
clients ||--o{ keywords : "analyzes"
backlink_campaigns ||--o{ backlink_prospects : "generates"
backlink_campaigns ||--o{ outreach_threads : "sends"
backlink_prospects ||--o{ outreach_threads : "interacts via"
backlink_prospects ||--o{ acquired_links : "resolves to"
backlink_campaigns ||--o{ acquired_links : "produces"
keyword_clusters ||--o{ keywords : "groups"
```

```
tenants {
    uuid id PK
    varchar slug UK
    varchar name
    tenant_plan plan
}
users {
    uuid id PK
    uuid tenant_id FK
    varchar external_id UK
    user_role role
}
clients {
    uuid id PK
    uuid tenant_id FK
    varchar domain
    client_status status
}
backlink_campaigns {
    uuid id PK
    uuid tenant_id FK
    uuid client_id FK
    campaign_type campaign_type
    campaign_status status
}
backlink_prospects {
    uuid id PK
    uuid tenant_id FK
```

```

        uuid campaign_id FK
        varchar domain
        prospect_status status
    }
    outreach_threads {
        uuid id PK
        uuid tenant_id FK
        uuid campaign_id FK
        uuid prospect_id FK
        thread_status status
    }
    acquired_links {
        uuid id PK
        uuid tenant_id FK
        uuid campaign_id FK
        uuid prospect_id FK
        link_status status
    }
}

```

7. BUSINESS MEMORY / TIME-SERIES TABLES

These tables act as the data foundation for business analytics, tracking historical trends, volatility, and pipeline changes.

7.1 Table: **campaign_health_snapshots**

Logs operational statistics for campaigns at set intervals to build trend lines.

- **Columns:** `id` (UUID PK), `tenant_id` (UUID), `campaign_id` (UUID FK), `health_score` (float), `acquisition_rate` (float), `reply_rate` (float), `progress` (float), `momentum` (float), `velocity` (float), `snapshot_data` (JSONB), `captured_at` (timestamp).
- **Indexes:** B-Tree index on `(campaign_id, captured_at)`.

7.2 Table: **keyword_metric_snapshots**

Tracks search engine position changes and volatility over time.

- **Columns:** `id` (UUID PK), `tenant_id`, `keyword_id` (UUID FK), `cluster_id` (UUID FK), `search_volume` (int), `difficulty` (float), `cpc` (float), `ranking_position` (int), `ranking_url` (varchar), `serp_features` (jsonb), `opportunity_score` (float), `captured_at`.

7.3 Table: `serp_volatility_snapshots`

Records macro fluctuations on SERPs by keyword and region.

- Columns: `id` (UUID PK), `tenant_id`, `keyword` (varchar), `geo` (varchar), `volatility_score` (float), `url_churn` (int), `position_changes` (int), `feature_changes` (jsonb), `top_10_domains` (jsonb), `captured_at`.

7.4 Table: `prospect_score_history`

Maintains records of how prospect values evolved during rescoring routines.

- Columns: `id`, `tenant_id`, `prospect_id` (FK), `campaign_id` (FK), `domain_authority`, `relevance_score`, `spam_score`, `composite_score`, `confidence`, `score_reason` (varchar), `captured_at`.

8. EVENT SOURCING PATTERN

The platform utilizes event sourcing inside `workflow_events` to construct the execution timeline and state of complex campaign workflows. When a campaign status transitions or user submits signals, the Event Sourcing engine:

- Validates the `sequence_number` matches the last known database sequence + 1.
- Injects the event data into `workflow_events`.
- Ingests the state update into the running campaign record in `backlink_campaigns`.

To rebuild a campaign's state at any historical moment, the engine queries `SELECT event_data FROM workflow_events WHERE stream_id = :campaign_id ORDER BY sequence_number ASC` and applies the mutations sequentially starting from state zero.

9. AUDIT INFRASTRUCTURE

The auditing subsystem is managed via `AuditLogMiddleware` on HTTP requests and explicit services for worker activity tracking.

Request/Worker API Call



AuditLogMiddleware / Service



Extract Tenant Context, IP, Actor (User/Agent)



Capture Before State (Snapshot query)



Process DB Mutation (SQL transaction)



Capture After State (Snapshot query)



Write to `audit\_log` Table (Append-Only)

- **Partitioning:** The table is partitioned by month using PostgreSQL `PARTITION BY RANGE (created_at)` to maintain quick query performance over millions of records.
- **State Capture:** `before_state` and `after_state` capture the JSON diffs of modified models.

10. DATABASE MIGRATION STRATEGY

Migrations are managed via **Alembic** under `backend/alembic/`.

- **Discovery:** All models are registered in `backend/src/seo_platform/models/__init__.py` to ensure `target_metadata` discovery.
- **Commands:**
 - Generate: `alembic revision --autogenerate -m "description"`
 - Execute: `alembic upgrade head`
 - Rollback: `alembic downgrade -1`

11. CRITICAL IMPLEMENTATION NOTES

11.1 asyncpg Native Enum Matching Issue

During Phase P2, it was discovered that **asyncpg** requires the text representations of Python enum values in queries to match the PostgreSQL native enum types **exactly**. If a python enum value is named `awaiting_approval` but the DB enum is defined as `AWAITING_APPROVAL`, query execution fails. All enum registration strings are therefore strictly standardized to **snake_case lowercase** globally.

11.2 Reusing Enum Types (create_type=False)

When defining enums in secondary models (e.g. `keyword_clusters` utilizing the `search_intent` enum defined in `keywords.py`), SQLAlchemy's default behavior is to attempt creation of the enum type again during migrations. To prevent migration errors, the `create_type=False` argument is applied to SQLAlchemy `Enum` columns where the type already exists.